

AI_USAGE_NOTE

Project Name:

NL-To-SQL Analytics Agent

AI Tools Used:

- * ChatGPT
- * Ollama (Llama 3.2)
- * Groq API

1. WHAT AI HELPED WITH DURING DEVELOPMENT

AI was used as a development assistant throughout the project lifecycle. The following areas were supported by AI:

PROJECT ARCHITECTURE

- * Assisted in designing the modular architecture of the application.
- * Suggested separation of functionalities into modules such as:
 - * agent_loop.py
 - * mcp_tools.py
 - * schema_tool.py
 - * sql_generator.py
 - * validator.py
 - * sql_executor.py
 - * chart_generator.py
 - * explanation_generator.py
 - * insight_generator.py

STREAMLIT DASHBOARD DEVELOPMENT

- * Assisted in building the Streamlit user interface.
- * Suggested layouts for query input, result display, dataset summary, manual SQL editor, and visualization sections.
- * Helped improve responsiveness and overall user experience.
- * Assisted in implementing provider selection for Ollama and Groq.

SQL GENERATION WORKFLOW

- * Assisted in designing the Natural Language to SQL workflow.
- * Helped create prompts for generating SQL queries using Ollama and Groq.
- * Assisted in implementing schema-aware SQL generation.
- * Suggested retry mechanisms through an Agent Loop workflow.

SQL VALIDATION AND SECURITY

- * Assisted in implementing validation rules to ensure safe query execution.
- * Suggested techniques for blocking unsafe SQL commands such as:
 - * DROP
 - * DELETE
 - * UPDATE
 - * INSERT
 - * ALTER
 - * TRUNCATE
 - * CREATE
 - * REPLACE
- * Assisted in preventing SQL injection and multi-statement execution.

AGENT LOOP IMPLEMENTATION

- * Assisted in designing an Agent Loop that:
- * Generates SQL
- * Validates SQL
- * Executes queries
- * Detects failures
- * Retries execution when required

MCP TOOL INTEGRATION

- * Assisted in creating MCP tools for schema access and future extensibility.
- * Suggested architecture for exposing database utilities through MCP.

DATA VISUALIZATION

- * Assisted in generating Plotly charts.
- * Suggested chart selection logic for:
- * Bar Charts
- * Line Charts
- * Pie Charts
- * Scatter Charts

TESTING AND DEBUGGING

- * Helped identify and fix coding issues.
- * Assisted in creating test cases for:
- * Schema extraction
- * SQL validation
- * SQL execution
- * Chart generation
- * Agent Loop functionality

2. CHALLENGES ENCOUNTERED WITH AI

SCHEMA HALLUCINATION

- * AI occasionally generated SQL queries using non-existent tables or columns when schema information was not provided explicitly.

CONTEXT ISSUES

- * When project files were renamed or modified, AI occasionally referenced outdated modules or functions.

SQL ACCURACY

- * Some generated SQL queries required manual verification and correction to match the actual database schema.

UI ADJUSTMENTS

- * Several iterations were required to achieve the final Streamlit layout and visualization appearance.

PROVIDER INTEGRATION

- * AI-generated integration examples occasionally required manual adjustments to correctly support both Ollama and Groq providers.

3. BEST PROMPTS USED

PROMPT 1

Generate a SQLite query for the user's question using only the provided database schema. Return only executable SQL.

PROMPT 2

Review the generated SQL query and determine whether it contains unsafe operations such as DROP, DELETE, UPDATE, INSERT, ALTER, CREATE, REPLACE, or multiple statements.

PROMPT 3

Create a responsive Streamlit dashboard that accepts natural language questions, displays generated SQL, query results, business insights, and Plotly visualizations.

PROMPT 4

Generate suitable Plotly visualizations automatically based on the structure of the query result dataset.

PROMPT 5

Implement an Agent Loop that generates SQL, validates it, executes it, retries on failure, and logs execution steps.

PROMPT 6

Review the entire repository and identify any missing deliverables, documentation, datasets, tests, or project requirements.

4. HUMAN CONTRIBUTION

Although AI assisted with code generation, debugging, testing, and documentation, all implementation decisions, integration work, testing, validation, and final review were performed manually.

The developer:

- * Implemented the final application architecture.
- * Integrated Ollama and Groq providers.
- * Developed the Streamlit dashboard.
- * Tested generated SQL queries.
- * Validated execution results.
- * Implemented manual SQL editing functionality.
- * Verified chart generation and export features.
- * Reviewed and corrected AI-generated outputs.
- * Ensured that the final application met project requirements.

CONCLUSION

AI was used as a development assistant to accelerate coding, debugging, testing, visualization, documentation, Agent Loop development, MCP integration, and SQL generation workflows.

Final implementation, verification, testing, and project integration were completed manually by the development team.

